

Luke Yarnall

Blacksburg, VA | LukeYarn8836@gmail.com | 703-485-7483 | lukeyarn8836-del.github.io

PROFESSIONAL SUMMARY

Mechanical Engineering student at Virginia Tech with hands-on manufacturing experience in metal additive manufacturing and quality control for aerospace and automotive clients including Lockheed, Boeing, and Sikorsky. Skilled in CNC operation, SOP documentation, root cause analysis, and CAD modeling, with a record of applying engineering principles in competitive and professional environments.

EDUCATION

Virginia Polytechnic Institute and State University

Bachelor of Science in Mechanical Engineering. GPA: 3.20.

Blacksburg, VA

Expected Spring 2027

Northern Virginia Community College (NOVA)

Associate of Science in Engineering. Magna Cum Laude. GPA: 3.682.

Annandale, VA

2024

EXPERIENCE

Metal 3D Printing Manufacturing Intern

June 2025 – July 2026

MELD Printworks

Christiansburg, VA

- Operated and maintained 3 metal additive manufacturing systems: one MELD L3 machine and two CD-14 deposition heads integrated onto refurbished CNC platforms, producing aerospace and automotive components.
- Monitored long-duration print jobs spanning 20 to 100 hours, including overnight shifts, to ensure calibration, environmental conditions, quality control, and safe shutdown procedures were maintained.
- Documented potential build failures and quality deviations in real time during active print cycles, ensuring accurate records for downstream root cause analysis and SOP compliance.
- Conducted root cause analysis on print failures and coordinated milling and resurfacing of affected plates.
- Supported production for customers including Lockheed, Boeing, Hyundai, IAI, and Sikorsky.

Printing and Manufacturing Lead

September 2024 – Present

3D-Printing Aircraft Competition Club (3D-PAC)

Blacksburg, VA

- Designed plane profiles and evaluated airframe aerodynamic stability using XFLR5 CFD to optimize flight performance.
- Modeled lightweight RC aircraft components and sub-assemblies in Fusion 360 and SolidWorks.
- Manufactured competition aircraft using Ender 5 Pro, Bambu P1S, CR-30 (infinite-axis), and Creality K2 Pro 3D printers.
- Repaired printing hardware, set production schedules, and managed print operations for national competition events.
- Earned 2nd place nationally in 2025 with MagicAirbus (14-sec flight) and 2nd in 2026 with HookieTail (22-sec flight).

SKILLS

Additive & Manufacturing: Metal Additive Manufacturing (MELD Friction Stir Process), FDM/SLA 3D Printing, CNC Machine Operation, Slicing (Bambu Studio, Cura, PrusaSlicer)

Quality & Analysis: Root Cause Failure Analysis (RCA), Standard Operating Procedures (SOPs), Quality Control & Inspection, XFLR5 (CFD)

CAD & Prototyping: SolidWorks, Fusion 360, NX Siemens, Onshape, Design for Manufacturability (DFM), Rapid Tooling Jigs & Fixtures

Programming & Tools: Python, MATLAB, Java, Microcontrollers (Arduino), Analog transmitters and receivers (RC controllers and video transmitters)

Languages: English (Native), Japanese (Fluent)

PROJECTS

Atlas Program (I-VI) <https://lukeyarn8836-del.github.io/#>

- Designed and built 3D printed RC planes for flight, video transmission, and payload delivery; portfolio on GitHub.

Swivel Sight

- Designed and fabricated a fully 3D printed firearm sight storage accessory using bolt hinges and bungee cord.

Battle Cups

- Built an Arduino-based device using continuous servos, an ultrasonic sensor, and switches to transport cups for a cup pong game.

AWARDS & HONORS

- VA Governor's Academic Excellence Award & Certificate of Recognition for Early College Scholars Program (2022)
- VA Board of Education Biliteracy Award (2022)
- LBSS Wrestling Team Academic Excellence Letter (2019-2022)
- NVCC Dean's List Honoree (Fall 2022)
- Nominated for the National Society of Leadership and Success Award (Spring 2023)